

A binary star system consists of two stars orbiting their common centre of mass. Star A has mass $M_A = 4 \times 10^{30}$ kg and Star B has mass $M_B = 1 \times 10^{30}$ kg. The straight-line distance between the two stars is $D = 5.0 \times 10^{12}$ m, and the gravitational constant is $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$. The two stars orbit their shared centre of mass in perfect uniform circular motion.

Assumptions

1. The stars are point masses; only their mutual gravitational attraction acts.
2. The orbits are circular and coplanar, about the common centre of mass.

Questions.

Q1. Equal and Opposite Forces. Compare the magnitude of the gravitational pull that Star A exerts on Star B with the pull that Star B exerts on Star A. State whether they are larger, smaller, or equal, and name the physics law that supports your conclusion.

Q2. The Zero-Force Point. Along the line connecting Star A and Star B there is a unique point where the total gravitational force on a tiny test mass m placed at that position cancels out to zero.

1. Let x be the distance from Star A to this zero-force point. Write the gravitational force-balance equation for the test mass.
2. Solve numerically to calculate the value of x .

Q3. Orbital Period. The zero-force point from part (b) is *not* the system's centre of mass. Both stars revolve around their common centre of mass with identical period T .

1. Write the equation equating the mutual gravitational force to the centripetal force for Star A.
2. Write the matching force equation for Star B.
3. Combine the two equations to derive the modified Kepler's Third Law for binary stars, then substitute the given numbers to solve for the orbital period T of the system.

Q4. Orbital Speeds. Derive and write a single equality relating M_A , M_B , v_A , and v_B . No numerical calculation is needed here.